SOLUTION ARCHITECTURE PROPOSAL

Kearney Intelligence Holarchy

A.T. Kearney Knowledge Governance & Adaptive AI Platform

Prepared by: Jeffrey Wallk, Senior Enterprise Architect & Knowledge Engineering Leader
The Value Enablement Group, LLC
April 2026 | Document Reference: KIH-ARCH-001-v1.0 | CONFIDENTIAL

SemanticGovernance	**HolonicArchitecture**	**TrustMetrology**	**NLP &Machine API**	**18-MonthRoadmap**

Executive Summary

The Kearney Intelligence Holarchy (KIH) is a structurally governed, trust-metrological knowledge architecture that transforms Kearney's intellectual capital from siloed repositories into a living, self-evolving holarchy of governed semantic assets. It is not an AI deployment — it is the architectural grammar through which every piece of Kearney's knowledge is made formally identifiable, composable, and metrologically trustworthy.

> Strategic Thesis: The firms that will dominate management consulting in the next decade are not those that deploy AI fastest — they are those that govern AI knowledge most rigorously. KIH moves Kearney from probabilistic AI assistance to deterministic knowledge orchestration: every insight delivered is structurally traceable, formally governed, and metrologically verified before it reaches a consultant or client.

Three Paradigm Shifts

From Data Management to Knowledge Governance	From Vector Search to Holarchic Retrieval	From Trust by Hopeto Trust by Architecture
Every piece of Kearney IP becomes a governed Knowledge Holon: formally identified, versioned, quality-validated, and semantically linked through persistent URI/IRI identity, ORSD governance contracts, and SHACL invariant enforcement.	Moving beyond flat RAG and vector databases, KIH traverses a four-level holarchy — Knowledge → Domain → Gateway → Enterprise — navigating governed meaning rather than retrieving probabilistic similarity matches.	Every insight carries a mathematically verifiable MTBH Trust Score stratified across five dimensions (WHO/WHAT/WHEN/WHERE/WHY). Hallucinations are not filtered — they are structurally prevented by the holarchy's governance DNA.

Commitment Delivery

- **Jarvis for Field Consultants & Sales:** live in Month 6, built on two production domain holons (Supply Chain + M&A/PE) with full RDSG Gateway and MTBH trust scoring.
- **Progressive Holarchy Scale-Out:** every subsequent phase adds domain holons without re-engineering the core. Complexity is managed architecturally, not operationally.
- **NLP Human Interfaces + Machine APIs:** deployed simultaneously from Month 6, enabling both human knowledge consumption and machine-to-machine semantic interoperability.
- **Measurement from Day 1:** MTBH dimensional trust metrics, SRE SLOs, and experiential outcome tracking active from first ingestion — not as a post-deployment reporting layer.

1. Architecture Overview — The Four-Level Holarchy

The KIH is organised around Arthur Koestler's holonic principle: every governed unit is simultaneously a self-contained whole (governed by its own internal invariants) and a composable part (interoperable within the larger system through shared governance protocols). This dual nature — autonomy and integration — is the structural property that makes the architecture both scalable and trustworthy.

Figure 1: Kearney Intelligence Holarchy — Four-Level Architecture Stack

1.1 The Four Holonic Levels

Level	Scope	Governance Mechanism	Kearney Application
L1 Knowledge Holon	Individual semantic asset (vocabulary, concept, SHACL shape, OWL class)	ORSD governance contract · KGCL change records · SHACL invariant enforcement · URI/IRI persistent identity · ISO 8000 quality dimensions	White papers, engagement frameworks, market data, methodology cards — each formally identified and governed
L2 Domain Holon	Component ontology + Domain Agent + Hybrid Vector DB + 6 PEPs	Domain ontology (OWL 2) · SHACL shapes · SWRL rules · Pinecone/Weaviate vector DB · MTBH dimensional score per holon	Supply Chain, M&A/PE, Digital, Operations, Sustainability — each a sovereign practice-area knowledge environment
L3 Gateway Holon	RDSG Semantic Gateway — holarchic traversal and composition protocol	RCB assembly · 6-stage enforcement · Circuit Breaker · MTBH 5D scoring · Simplex Action Gate · Sanitization anti-collapse	All Jarvis interfaces route through the gateway — the single point of semantic authority assembly and trust verification
L4 Enterprise Holarchy	Full federated governance ecosystem across all practices and lifecycle stages	Enterprise Semantic Governance Council · CoE · Value Stream Ontology · Continuous engineering loop · SRE management	Global strategic intelligence synthesis, cross-practice pattern detection, executive macro-trend orchestration

1.2 The Five Holonic Design Properties

Three properties — derived from Koestler's original formulation and extended by Kurt Cagle's holonic graph theory — must be present in every component of the KIH:

- **Autonomy:** Each holon governs itself according to its own SHACL invariants without requiring continuous external instruction. A domain holon can operate in island mode when higher-level holonic connectivity is degraded.
- **Integration:** Each holon exposes standardised PEP-defined interfaces that allow it to compose with other holons through governance protocols rather than shared schemas. Adding a new practice area holon never requires re-engineering the core.
- **Identity:** Each holon carries a persistent URI/IRI, an ORSD governance contract, and a KGCL version history that travels with it across all contexts and lifecycle stages.
- **Resilience through Semi-Autonomy:** Distributed governance across component holons prevents single-point-of-failure schema collapse. A change in the Sustainability domain ontology does not require global re-engineering.
- **Digital Orchestration over Documentation:** The holarchy governs what should happen and ensures operational conformance — it does not merely record what happened. This is the structural definition of enterprise-grade AI trustworthiness.

2. Supply Chain Domain Holon — Architecture & Knowledge Services

The Supply Chain Domain Holon is the first production domain in Phase 1 and serves as the architectural reference implementation. Every subsequent practice area holon replicates this pattern with domain-specific vocabularies, ontologies, agents, and PEP configurations — without modifying the core architecture.

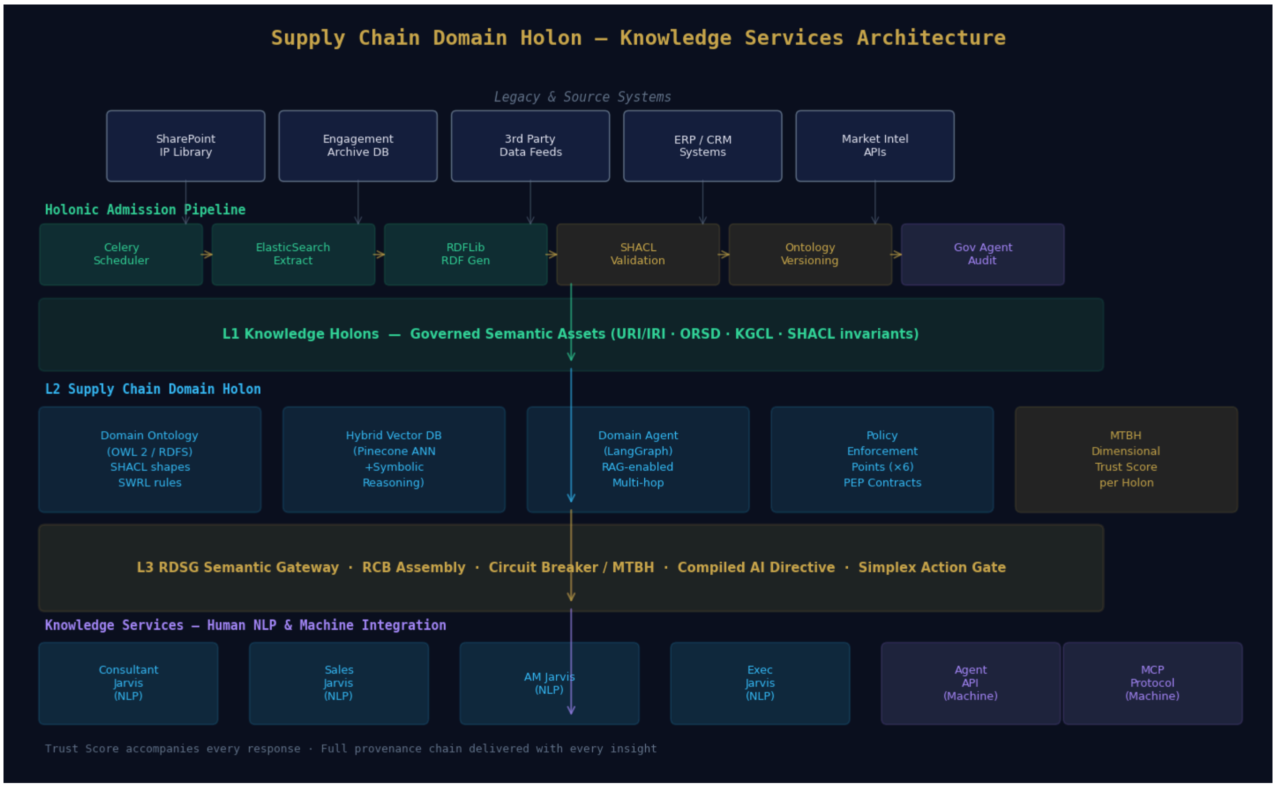

Figure 2: Supply Chain Domain Holon — Knowledge Services from Source Systems to Jarvis Interfaces

2.1 Domain Holon Components

Each domain holon is a sovereign, self-governing knowledge environment containing five integrated components:

Component	Technology	Holonic Function
Domain Ontology	OWL 2 (W3C) · RDFS · SKOS · OMG ODM scaffold	Defines the semantic vocabulary, class hierarchy, and formal relationships for the practice area. SHACL shapes enforce what valid instances look like. SWRL rules govern temporal and inferential reasoning.
Domain Agent	LangGraph / AutoGen orchestration · Anthropic Claude (governed)	Receives Compiled AI Directives from L3 Gateway — never raw prompts. Executes within holonic governance boundaries. MTBH per-agent dimensional scoring monitors agent fidelity continuously.
Hybrid Vector DB	Pinecone / Weaviate (ANN + symbolic)	Level 1: ANN searches for semantic similarity within the domain ontology index space. Level 2: SPARQL / SHACL validation confirms retrieved assets satisfy domain invariants. Prevents cross-domain contamination.

Component	Technology	Holonic Function
Policy EnforcementPoints (×6)	Custom PEP framework · RDSG enforcement engine	Compliance · Risk · Legal · Security · Data Governance · AI Governance. Each PEP is evaluated at L3 Gateway Steps 1–4 before the compiled AI Directive is assembled. PEPs are holarchic contracts, not filters.
MTBH Monitor	Circuit Breaker engine · KGCL telemetry	Dimensional trust scoring (WHO/WHAT/WHEN/WHERE/WHY) per agent invocation. Targeted remediation without systemic shutdown. Trip events quarantine outputs; sanitization layer prevents training contamination.

2.2 Holonic Admission Pipeline — Legacy System Integration

The admission pipeline is the mechanism by which Kearney's existing repositories (SharePoint, engagement databases, third-party data feeds, ERP/CRM systems) are onboarded into the holarchy as governed Knowledge Holons. This is not data migration — it is holonic onboarding. Every asset must satisfy the domain holon's SHACL invariants before it can compose with the governance structure.

- **Step 1 — Request Scheduler (Celery + Redis):** Queue management for the holonic admission process. Every inbound asset is assigned a tracking ID and monitored through the admission lifecycle.
- **Step 2 — Asset Extraction (ElasticSearch + Python):** Asset discovery across SharePoint, engagement archives, and external APIs. Schema, version, datetime stamp, author, and engagement context extracted.
- **Step 3 — RDF Generation (RDFLib):** Asset transformed into an RDF triple structure conformant with the target domain ontology template. URI/IRI identity assigned.
- **Step 4 — SHACL Validation (Metaphactory):** Holonic admission test. Does the candidate's Knowledge Holon satisfy the domain's SHACL invariants? Failures are quarantined with KGCL rejection records.
- **Step 5 — Ontology Versioning:** Domain ontology version incremented. KGCL change records are generated with full WHO/WHAT/WHEN/WHERE/WHY metadata for every admission event.
- **Step 6 — Governance Agent Audit:** Policy conformance verification: metadata governance, naming conventions, provenance chain, and audit trail validated before PRD environment admission.

> DEV / STG / PRD Holonic Lifecycle: DEV is holonic construction (agent training, ontology authoring, SHACL validation). STG is a holonic composition (Top-Down core ontology reconciled with Bottom-Up legacy schemas via R2RML). PRD is holonic production (governed by semantic search, conversational Jarvis, agent federation, and real-time trust scoring).

3. RDSG Gateway Holon — Six-Stage Semantic Traversal

The RDSG (Requirements-Driven Semantic Gateway) is the architectural heart of the KIH — the Level 3 holarchic protocol through which all knowledge services are delivered. It is not a pipeline that transforms data; it is a governance composition mechanism that assembles semantic authority from every domain holon it touches and delivers it as a behavioural directive to the AI agent.

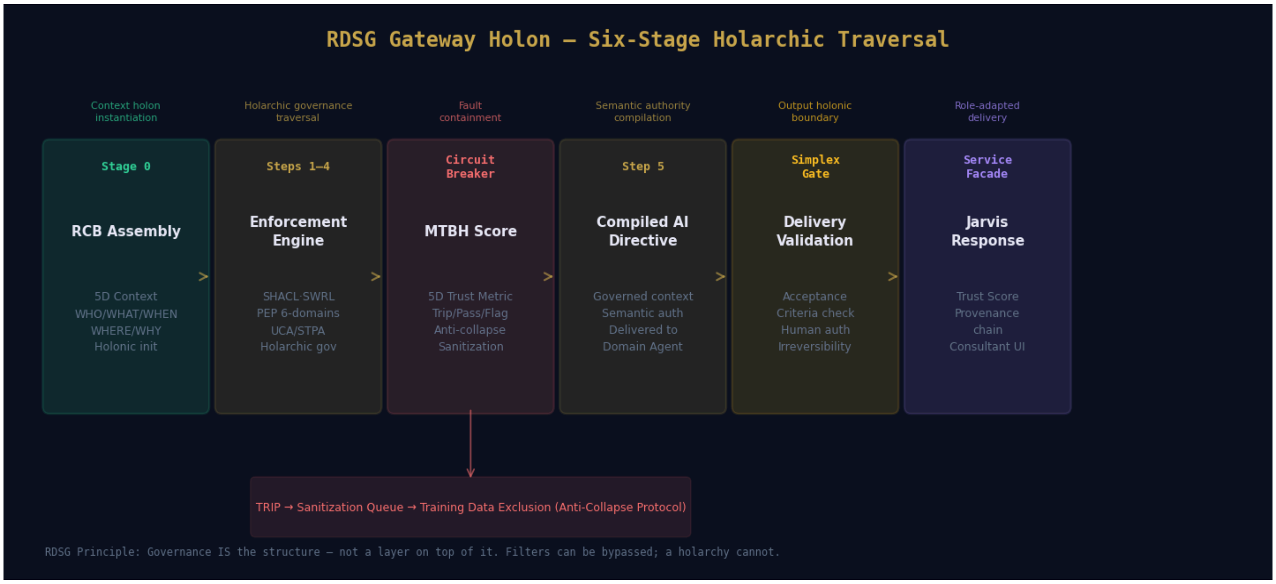

Figure 3: RDSG Gateway Holon — Six-Stage Holarchic Traversal with Circuit Breaker

3.1 The Six-Stage Flow

Stage	Name	Function	Trust Mechanism
Stage 0	**RCB Assembly**	The Structured Prompt Object is decomposed into five relevancy dimensions (WHO/WHAT/WHEN/WHERE/WHY). Appropriate domain holons are activated. The Requirement Context Bundle is the initial holonic context instantiation — not an input packet.	Persona mapping, domain routing, intent-tag governance, and temporal context establishment
Steps 1–4	**Enforcement Engine**	Four sequential governance layers: (1) SHACL constraint validation against activated domain holons; (2) SWRL temporal rule application; (3) PEP compliance evaluation across 6 governance domains; (4) UCA safety coverage verification (STPA/STAMP).	SHACL invariants, SWRL temporal reasoning, PEP Compliance/Risk/Legal/Security/DataGov/AIGov, UCA boundary enforcement
Circuit Breaker	**MTBH Scoring**	Dimensional trust scoring across WHO/WHAT/WHEN/WHERE/WHY. HARD_TRIP for irreversible actions (human auth required). SOFT_TRIP for low-confidence routing (reroute + flag). WARN for borderline cases (deliver with caveat). All trips are quarantined for sanitization.	MTBH 5D score, Intent Delta, Value-Drift Coefficient, Anti-collapse sanitization queue

Stage	Name	Function	Trust Mechanism
Step 5	**Compiled AI Directive**	Governance outputs from all four enforcement holons are synthesised into a behavioural directive. The agent does not receive instructions — it receives governed context carrying domain-specific semantic authority from every holon it touches.	Full provenance chain embedded in directive; agent operates within holonic governance boundaries
Simplex Gate	**Delivery Validation**	Agent response validated against AcceptanceCriteria. Irreversible actions are suspended pending human authorisation. STPA UCA boundary enforced. Output holonic boundary.	AcceptanceCriteria validation, human-in-the-loop gate, irreversibility detection, final MTBH check
Service Facade	**Jarvis Response**	Role-adapted response delivered to appropriate interface (Consultant/AM/Sales/Executive Jarvis or Machine API). MTBH Trust Score and provenance chain accompany every insight delivered.	Trust Score displayed with every response; provenance traceable to specific IP assets and holonic governance decisions

3.2 The Five Relevancy Dimensions in Kearney Context

Every query traversing the KIH Gateway is decomposed into five holonic context vectors that govern routing, retrieval, compliance, and trust measurement simultaneously. These are not query metadata fields — they are the governance axes of the holarchy.

Dim.	Holonic Boundary	Kearney Application	MTBH Metric
WHO	Persona-holon: identity context of the initiating agent or user	Consultant vs. AM vs. Sales vs. Executive — governs IP access scope, PEP permissions, and role-aware response format. Prevents a sales context from accessing client-confidential engagement data scoped to delivery teams.	MTBH_WHO: Persona routing fidelity ≥98%
WHAT	Concept-holon: the specific knowledge artifact being accessed	The specific framework, methodology, or IP asset queried — OWL class resolution against the Kearney core ontology. Ensures the correct supply chain framework is retrieved, not a superficially similar operations model.	MTBH_WHAT: IP concept resolution ≥99%
WHEN	Temporal-holon: the time context of the interaction	Engagement recency, white paper version, market data timestamp — SWRL temporal rules enforce currency of insights. White papers older than 24 months are flagged unless explicitly requested for historical context.	MTBH_WHEN: Temporal anchoring ≥95%
WHERE	Domain-holon: the specific component ontology boundary	Which practice area ontology to activate? Prevents cross-domain hallucination: a supply chain pricing model must not contaminate an M&A valuation query even if semantic similarity scores suggest relevance.	MTBH_WHERE: Domain routing ≥97%
WHY	Intent-holon: the governance purpose of the interaction	Client-facing vs. internal; proposal vs. research; executive briefing vs. consultant hypothesis. Activates appropriate PEP compliance rules and IP licensing constraints for each purpose.	MTBH_WHY: Intent-governance ≥96%

4. Knowledge Services — NLP for Humans & Machine Integration

The KIH delivers knowledge through two parallel service channels from the same governed holarchy: natural language interfaces for human consumers (the four Jarvis roles) and structured API/protocol interfaces for machine consumers (agents, services, and downstream processes). Both channels receive the same MTBH-verified, provenance-traced governed knowledge — differentiated only by delivery format and interface protocol.

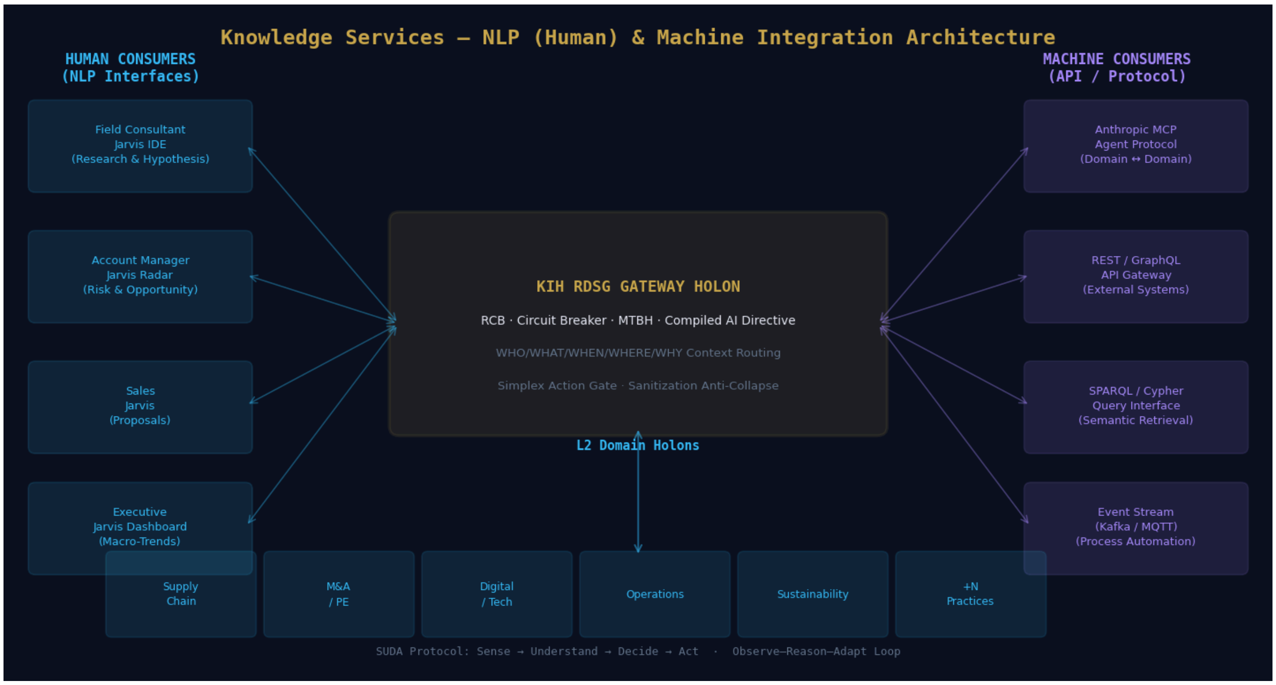

Figure 4: Knowledge Services — NLP (Human) & Machine Integration Architecture

4.1 Human NLP Interfaces — The Four Jarvis Roles

Jarvis Role	Holonic Level	Core Capabilities	Experiential Outcome Targets
Field Consultant Jarvis	L2/L3 Domain + Gateway	IDE-like research companion · Hypothesis pressure-testing against historical engagement data · Analytical model auto-completion · Presentation module drafting · Trust Score per claim · Provenance chain to source IP	Research time reduction ≥60% · Hypothesis accuracy ≥85% · NPS ≥40 from pilot cohort
Account Manager Jarvis	L4 Enterprise Holarchy	Proactive client risk & opportunity radar · Third-party signal enrichment · Talking point generation grounded in latest white papers · Client portfolio holon integration · Real-time alerting with MTBH confidence	Client risk alert accuracy ≥90% · Engagement renewal lead-time improvement · Zero unattributed claims
Sales Jarvis	L4 Cross-Domain	Automated proposal generator · Cross-domain pain-point-to-partner-expertise matching · Pitch deck construction from proven IP · Win/loss pattern integration · Competitive intelligence synthesis	Proposal generation time ≥65% faster · Proposal advance rate ≥35% · 100% IP provenance on all proposals

Jarvis Role	Holonic Level	Core Capabilities	Experiential Outcome Targets
Executive Jarvis	L4 Enterprise Holarchy	Macro-trend dashboard · Cross-practice pattern detection · Strategic hiring and practice development signals · Scenario modelling across value stream ontology · Quarterly holarchic evolution report	Strategic trend detection lead-time +3 months · Cross-practice IP utilisation ≥75% · Knowledge monetisation index growth

4.2 Machine Integration — Emergent Capabilities for Processes, Services & Agents

The same holarchy that powers human Jarvis interfaces exposes structured machine-integration endpoints for automated processes, enterprise services, and agent-to-agent communication. Machine consumers receive governed knowledge through protocol-defined interfaces — the same MTBH-verified, provenance-traced outputs, formatted for programmatic consumption.

Integration Layer	Protocol / Technology	Machine Consumer Use Cases	Holonic Governance Mechanism
Anthropic MCP(Model Context Protocol)	MCP (Anthropic standard)Domain-to-domain agent communication	Agent federation across practice area holons · Cross-domain reasoning workflows · Autonomous multi-hop knowledge traversal · Agentic process orchestration	MCP is the holarchic agent communication protocol at L2/L3 — standardised, auditable, zero-trust access controls with agent registry tracking and MTBH per-agent scoring
REST / GraphQL API Gateway	REST JSON-LD · GraphQLOpenAPI 3.0 schema	Enterprise system integration (ERP/CRM) · Dashboard data feeds · Third-party analyst platform integration · Client portal knowledge delivery · Partner API access with IP licensing enforcement	API Gateway sits behind L3 RDSG Gateway — all calls pass through RCB assembly, PEP enforcement, and MTBH scoring. JSON-LD responses carry semantic context and provenance metadata.
SPARQL / CypherSemantic Query Interface	SPARQL 1.1 (W3C) · CypherGraphDB / Neo4j endpoint	Multi-hop relationship queries across knowledge graph · Ontology-level pattern analysis · Compliance audit queries · Provenance chain traversal · Impact analysis for IP change management	SPARQL queries execute against the domain holon graph layer — governed by SHACL constraints and SWRL rules. Query results include provenance metadata and MTBH confidence scores.
Event Stream(Process Automation)	Apache Kafka · MQTT Event-driven architecture	Automated compliance monitoring · Real-time market signal processing · Supply chain event-triggered knowledge retrieval · Proactive AM alert generation · Continuous ontology enrichment triggers	Event consumers receive sanitized, MTBH-validated knowledge payloads. Circuit Breaker outputs are excluded from event streams. KGCL change events trigger downstream consumers automatically.

4.3 SUDA — Holonic Traversal Protocol for All Interactions

Every interaction with the KIH — whether from a human consultant typing a query or a downstream agent submitting a SPARQL request — follows the SUDA holonic traversal protocol:

- **Sense:** Collect information about the input or prior conversation context. WHO/WHAT/WHEN/WHERE/WHY dimensional tagging of the interaction builds the relevancy context vector that guides all subsequent holonic routing.

- **Understand:** Use the knowledge graph to infer relationships and connect entities. L2 domain holon activation — the appropriate domain ontology is invoked, SWRL rules evaluated, and semantic meaning resolved within its governance context.
- **Decide:** Identify which parts of the knowledge graph are most relevant. L3 Gateway traversal — RCB assembled, PEPs evaluated, and compiled AI Directive built. This is governance in action: not retrieval, but semantic authority assembly.
- **Act:** Pass the governed context to the AI agent or machine consumer to enhance the response with relevant, trusted insights. The agent operates within holonic governance boundaries established in the preceding phases.

5. Transformation Roadmap — 18-Month Holarchic Build

The roadmap is structured around four progressive phases, each delivering a complete functional increment of the holarchy. No phase is merely a technical building block waiting for future payoff — each delivers measurable business value. The holonic architecture is the enabling property: because domain holons are sovereign and composable, each phase extends the holarchy without re-engineering its foundation.

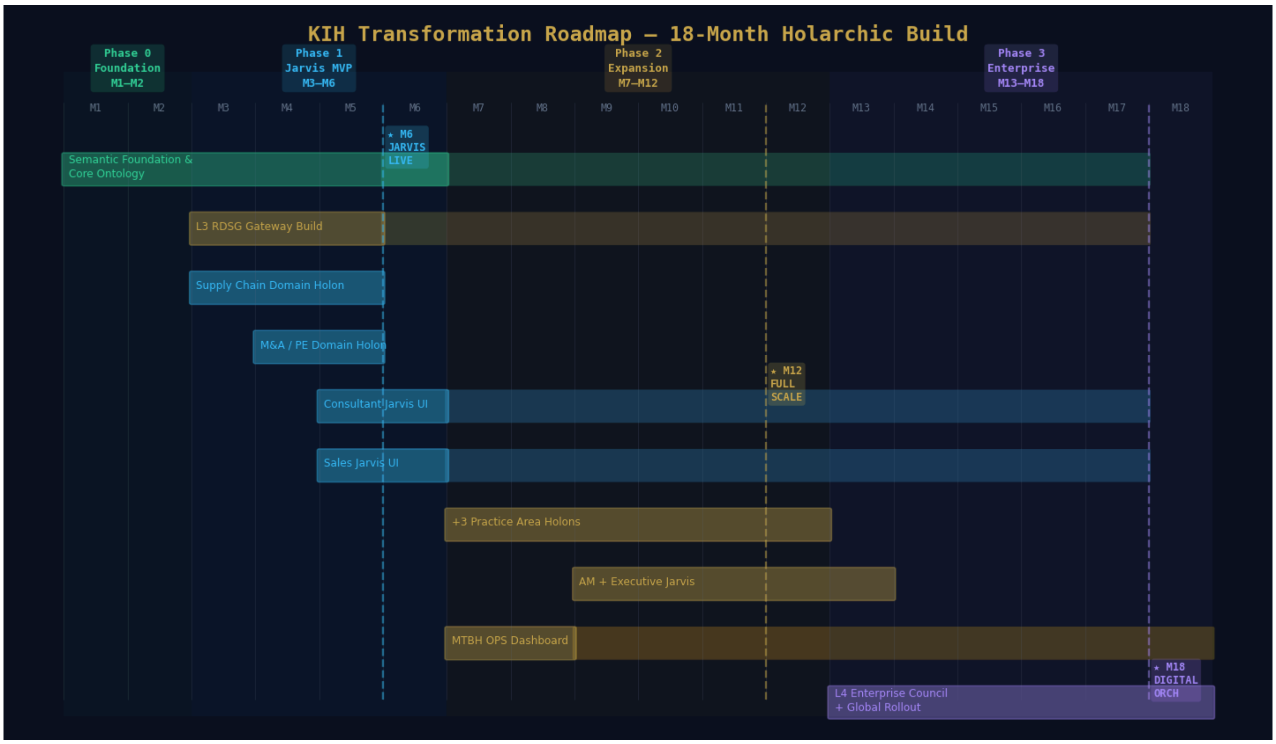

Figure 5: KIH Transformation Roadmap — 18-Month Holarchic Build with Key Milestones

Phase 0: Semantic Foundation (Months 1–2)

OBJECTIVE: STRUCTURE THE UNSTRUCTURED. ESTABLISH THE HOLONIC GRAMMAR BEFORE ANY AI IS DEPLOYED.

- **IP Audit & Cataloguing:** Comprehensive audit of all SharePoint repositories, engagement databases, proprietary systems, and third-party data contracts. Every asset assigned a URI/IRI identity — the prerequisite for holonic admission.
- **Kearney Core Ontology:** RDF/OWL enterprise-level shared vocabulary and polymorphic inheritance protocol. Defines the shared governance grammar that all domain holons must conform to — not the schema they are constrained by.
- **ORSD Template Library:** Governance contract specifications for all six core IP asset types (white papers, frameworks, engagement data, market intel, taxonomies, methodology cards).
- **KGCL Protocol Activation:** Change records with full WHO/WHAT/WHEN/WHERE/WHY metadata capture for every IP modification. Audit trail architecture is live from Day 1.
- **ISO 8000 Quality Mapping:** Eight data quality dimensions mapped to SHACL constraints for the inbound admission pipeline. Measurement is active before Jarvis is built.
- **Baseline SRE Metrics:** Established for all existing AI tools currently in use at Kearney. Provides the before-state benchmark for all subsequent value measurement.

> Phase 0 Governance Gate (Month 2): Core ontology validated by Enterprise Semantic Governance Council · ≥500 IP assets admitted as governed Knowledge Holons with SHACL validation ≥98% · ORSD templates approved for all six IP asset types · KGCL audit trail demonstrating full 5D metadata capture.

Phase 1: Jarvis MVP — GO LIVE Month 6 (Months 3–6)

OBJECTIVE: DELIVER JARVIS TO FIELD CONSULTANTS AND SALES. TWO PRODUCTION DOMAIN HOLONS. FULL RDSG GATEWAY WITH MTBH TRUST SCORING.

- **Month 3 — Supply Chain Domain Holon:** OWL 2 ontology, SHACL shapes, SWRL temporal rules, domain agent (LangGraph + Claude), Pinecone vector DB indexed against domain ontology, 6 PEPs configured and tested.
- **Month 4 — M&A / PE Domain Holon:** Identical stack to Supply Chain with practice-specific vocabularies and FIBO-grounded financial process ontology. A cross-domain alignment protocol was established between SC and M&A holons.
- **Month 4–5 — L3 RDSG Gateway:** Operational: RCB assembly, 6-stage enforcement engine, Circuit Breaker with MTBH 5D scoring, Simplex Action Gate, sanitization queue, human-in-the-loop authorization protocol.
- **Month 5 — Field Consultant Jarvis:** IDE-like interface with hypothesis pressure-testing, Trust Score display per response, full provenance chain. Analytical model auto-completion grounded in the Supply Chain and M&A domain holons.
- **Month 5 — Sales Jarvis:** Cross-domain proposal generator — matches client pain points to partner expertise holons. Pitch deck construction from verified IP. Win/loss patterns feed back into the value stream ontology.
- **Month 6 — Pilot Launch:** 20–30 consultants, 5–10 sales personnel. MTBH dashboard live. Structured experiential feedback loops are activated. Governance gate evaluation.

> Month 6 Go-Live Success Criteria: MTBH_WHAT ≥95% on IP concept resolution · P95 Consultant Jarvis latency ≤4 seconds · 100% of responses carry provenance chain · Circuit Breaker trip rate ≤2% · NPS ≥40 from pilot cohort · Sales proposal generation time reduced by ≥60%.

Phase 2: Holarchy Expansion (Months 7–12)

OBJECTIVE: ALL PRACTICE AREAS ARE PRODUCTION DOMAIN HOLONS. AM, AND EXECUTIVE JARVIS LIVE. FULL MTBH OBSERVABILITY.

- **Domain Holon Scale-Out (M7–M11):** Digital/Technology (M7), Operations (M8), Sustainability (M9), Healthcare/Pharma (M10), Public Sector/Defense (M11). Each holon added using the Phase 1 reference architecture — no core re-engineering required.
- **AM Jarvis (Month 9):** Proactive client risk & opportunity radar with third-party signal integration. Client portfolio holon personalises every alert and recommendation. MTBH_WHY monitors intent-governance alignment.
- **Executive Jarvis (Month 10):** Macro-trend dashboard synthesising cross-practice pattern detection across the full holarchy. First capability requiring L4 Enterprise Holarchy governance engagement.
- **MTBH OPS Workbench (Month 7):** Tom Sawyer Perspectives dashboard live — holarchic observatory across all four levels. Value-Drift Coefficient monitoring (Month 8), Intent Delta automated detection (Month 9).
- **Cross-Domain Value Stream Ontology (Month 12):** Enterprise-level holonic performance model linking all practice holons. Enables cross-practice engagement intelligence — insights impossible in any single practice area.
- **Global Office Rollout Begins (Month 12):** Multi-language support. Regional ontology extensions are admitted through the holonic admission protocol.

Phase 2 Compounding Advantage: Each new domain holon adds exponential connectivity to the holarchy, not linear complexity. Cross-domain traversal across Supply Chain + M&A + Digital holons produces insights that would require three separate expert teams in a conventional architecture.

Phase 3: Enterprise Holarchy & Digital Orchestration (Months 13–18)

OBJECTIVE: THE HOLARCHY GOVERNS WHAT SHOULD HAPPEN, NOT MERELY RECORDS WHAT HAPPENED. GLOBAL ROLLOUT. SELF-EVOLVING ARCHITECTURE.

- **Enterprise Semantic Governance Council (M13):** Formally constituted L4 governance authority. Cross-practice conflict resolution protocol operational. LKIF deontic norms integrated for compliance-as-code governance.
- **Global Deployment (M14):** All regional offices. Full SRE management of AI infrastructure. Automated failover and real-time degradation monitoring. Graceful degradation to pre-compiled L1/L2 rules when the L3 gateway is stressed.
- **Self-Evolving Holarchy (M15):** Observe–Reason–Adapt loop autonomous: RAG-discovered entities validated by Domain Governance Teams and admitted via KGCL protocol. Design-time governance nourished continuously by run-time operational intelligence.
- **Practice-Area Bespoke Extensions (M16):** M&A deal analytics agent, Supply Chain risk prediction holon, Digital transformation pattern library. Custom agents built on the KIH core architecture via documented extension protocol.
- **Third-Party Ecosystem Integration (M17):** Financial data feeds, analyst report networks, regulatory monitoring services — all governed by SHACL data contracts and temporal SWRL rules for currency enforcement.
- **Digital Orchestration Operational (M18):** The KIH governs Kearney's competitive intelligence strategy in real time. The holarchy is the strategy engine — the distinction between "operating the system" and "modelling the system" disappears.

6. Measurement & Trust Metrology Framework

Measurement in the KIH is not reporting — it is holarchic health monitoring. Every metric maps to a specific holonic level, a specific governance pillar, and a specific remediation protocol. The MTBH (Mean Time Between Hallucinations) dimensional framework is the primary trust instrument: five stratified metrics that localise failure to its holonic origin and enable targeted remediation without systemic shutdown.

> MTBH Design Principle: MTBH must never be operated as an aggregate score. A domain-routing failure (MTBH_WHERE) must not trigger suspension of concept resolution (MTBH_WHAT) in an unrelated domain holon. Dimensional stratification is the mechanism that makes targeted remediation possible — not an optional reporting enhancement.

6.1 MTBH Dimensional Trust Metrics

Dimension	Holonic Boundary	Kearney Application	Target	Remediation
MTBH_WHO	Persona-holon: identity context routing fidelity	Consultant vs. AM vs. Executive context isolation · IP access scope enforcement · Role-aware response formatting	**≥98%**	Persona PEP revalidation · Agent registry audit
MTBH_WHAT	Concept-holon: IP knowledge artifact resolution accuracy	Correct framework / methodology identified from Kearney knowledge graph · OWL class resolution precision	**≥99%**	SHACL constraint review · Domain ontology enrichment
MTBH_WHEN	Temporal-holon: data currency and anchoring fidelity	Market data freshness enforcement · White paper version currency · SWRL temporal rule activation	**≥95%**	Temporal rule update · Force re-ingestion of stale assets
MTBH_WHERE	Domain-holon: practice area boundary routing fidelity	Supply chain IP not contaminating M&A responses · Cross-domain hallucination prevention · Vector DB partition integrity	**≥97%**	Context Router Agent retraining · WHERE PEP boundary review
MTBH_WHY	Intent-holon: governance purpose alignment fidelity	Client-facing vs. internal output correctly governed · Proposal vs. research PEP activation · IP licensing constraint enforcement	**≥96%**	Intent-tag recalibration · Contextual Relevancy Agent retuning

6.2 SRE Service Level Objectives

SLO Category	Metric	Target	Alert & Remediation Protocol
AI Accuracy	Precision/Recall on IP retrieval	**P≥0.95R≥0.90**	Circuit Breaker soft trip · Domain ontology gap analysis
Trust Fidelity	MTBH aggregate (all 5 dimensions)	**≥95% per dim.**	Dimensional degradation alert → Domain Governance Team within 2hrs
Insight Latency	P95 Jarvis gateway response time	**≤4sConsultant**	Domain agent horizontal scaling · Gateway performance tuning

SLO Category	Metric	Target	Alert & Remediation Protocol
Data Currency	Stale asset rate (>90 days)	**≤5%active holons**	MTBH_WHEN degradation · Forced re-ingestion or deprecation flagging
Catalog Health	SHACL validation pass rate (inbound)	**≥98%**	Ingestion Rejection Rate spike → source data quality review
Agent Uptime	Domain agent availability per holon	**≥99.5%**	Graceful degradation to pre-compiled L1/L2 rules during outage
Anti-Collapse	Training contamination rate (tripped outputs)	**0%**	Alert on any non-zero value · Immediate sanitization queue audit
Value Drift	Value-Drift Coefficient per agent	**≤0.15**	Domain Governance Team review initiated · Agent objective realignment

Appendix A: Acronyms & Definitions

All acronyms and technical terms used in this document are defined below in alphabetical order.

Acronym / Term	Definition & Architecture Role
ANN	Approximate Nearest Neighbour — the vector similarity search algorithm used in Level 1 of the KIH hybrid retrieval strategy within domain holon vector databases (Pinecone, Weaviate). Provides fast semantic similarity matching indexed against domain ontology classes.
CB	Circuit Breaker — the holonic fault containment protocol within the L3 RDSG Gateway that evaluates MTBH dimensional scores and routes interactions to HARD_TRIP, SOFT_TRIP, or WARN states. Tripped outputs are excluded from agent training data via the sanitization anti-collapse layer.
CoE	Center of Excellence — the Semantic Governance CoE operating at L1/L3 of the KIH governance hierarchy, responsible for platform operations, cross-domain integration, agent development, and training and enablement.
DCAT	Data Catalog Vocabulary (W3C) — the knowledge holon catalog metadata standard used for PRD catalog publishing and vector DB metadata structure at L1/L2 of the holarchy.
DEV	Development Environment — the holonic construction lifecycle stage where domain holons are built and tested: vendor tool management, agent development, and inbound/processing/outbound service construction.
Digital Orchestration	Kurt Cagle's term for a knowledge system that governs what should happen and ensures operational conformance — as distinct from Digital Documentation, which merely records what happened. The strategic outcome of the KIH at Month 18.
DOL	Distributed Ontology, Model and Specification Language (OMG) — the cross-holonic interoperability standard supporting heterogeneous model integration at L2/L4 cross-domain boundaries.
DPROD	Data Products Ontology (OMG) — the data product holon description standard governing data product metadata, entitlements, and lineage for search and discovery at L2 domain holons.
FIBO	Financial Industry Business Ontology (EDM Council) — the reference ontology for the Finance Domain Holon, providing FP&A process/measurement/legal constraint classes grounded in industry-standard financial semantics.
GraphRAG	Retrieval-Augmented Generation over Knowledge Graphs — the KIH retrieval strategy that traverses multi-hop relationships in the holarchic knowledge graph (e.g., "How did this pricing framework affect an EU client in the past 3 years?") rather than performing flat vector similarity search.
Holon	An entity that is simultaneously a self-contained whole (governed by its own internal invariants) and a composable part of a larger system (interoperable through shared governance protocols). Introduced by Arthur Koestler in The Ghost in the Machine (1967).
Holarchy	The nested organisational structure of holons governed by protocol rather than centralised hierarchy. In the KIH: L1 Knowledge Holons → L2 Domain Holons → L3 Gateway Holon → L4 Enterprise Holarchy.
Intent Delta	The measured discrepancy between the AI agent's logical understanding of a task and the physical/operational reality it is operating within. A holonic misalignment metric monitored by the L3 Circuit Breaker.
IP	Intellectual Property — Kearney's body of white papers, engagement frameworks, methodologies, case data, and analytical models. The raw material of the L1 Knowledge Holon layer.
IRI	Internationalised Resource Identifier — the persistent holonic identity assigned to every Knowledge Holon. The IRI travels with the holon across all contexts and lifecycle stages.
ISO 8000	International Standard for Data Quality — the eight quality dimensions (Accuracy, Completeness, Consistency, Timeliness, Validity, Uniqueness, Relevance, Traceability) mapped to SHACL constraints as holonic invariants at L1 of the KIH.

Acronym / Term	Definition & Architecture Role
KGCL	Knowledge Graph Change Language — the run-time change record protocol for knowledge holons. Every change (add, update, delete) is expressed as a KGCL record with WHO/WHAT/WHEN/WHERE/WHY metadata, version information, business rationale, and conflict resolution policy.
KIH	Kearney Intelligence Holarchy — the full four-level holonic semantic governance and AI knowledge architecture described in this proposal.
KSWG	Knowledge Systems Working Group (INCOSE) — the professional working group co-led by the author, whose "Context is King" framework informs the core/engineering holarchy topology of the KIH.
LKIF	Legal Knowledge Interchange Format — the deontic norm specification language used to express governance obligations (must), permissions (may), and prohibitions (must not) as first-class ontological properties of knowledge holons. Enables compliance-as-code governance.
MCP	Model Context Protocol (Anthropic) — the holarchic agent communication protocol at L2/L3, enabling standardised, auditable, zero-trust communication between domain agent holons and the gateway.
MTBH	Mean Time Between Hallucinations — the primary holonic health metric of the KIH. Measures how long the holarchy's governance DNA remains intact between AI fidelity failures, stratified across five dimensions (WHO/WHAT/WHEN/WHERE/WHY). Proposed by the author as a formal trust metric for AI governance systems.
NLP	Natural Language Processing — the interface modality of the four Jarvis human consumer interfaces. Consultants, AMs, Sales, and Executives interact with the KIH in natural language; the RDSG Gateway mediates between natural language queries and the holarchic knowledge graph.
ODM	Ontology Definition Metamodel (OMG) — the holonic ontology design standard providing UML/ER to OWL/RDF mapping at L1/L2.
ORSD	Ontology Requirements Specification Document — the design-time specification of a knowledge holon's purpose, scope, and governance requirements. Establishes what the holon should be before it is instantiated. The holonic equivalent of a system requirements baseline in MBSE.
OWL	Web Ontology Language (W3C OWL 2) — the knowledge holon semantic definition language. Core, domain, and mapping ontologies in the KIH are expressed in OWL 2.
PEP	Policy Enforcement Point — the formal holarchic contract defining how a domain holon composes with the L3 Gateway. Six PEP governance domains: Compliance, Risk, Legal, Security, Data Governance, AI Governance. Evaluated at RDSG Steps 1–4.
PRD	Production Environment — the holonic production lifecycle stage where composed domain holons are published for downstream consumption. The outbound service portfolio (semantic search, conversational Jarvis, agent federation, event streams) operates in PRD.
RAG	Retrieval-Augmented Generation — the mechanism by which the KIH extends itself. In the holonic architecture, RAG is not merely a retrieval technique but a self-evolution protocol: RAG-discovered entities pass through Domain Governance Team review and KGCL admission before becoming permanent knowledge holons.
RCB	Requirement Context Bundle — the structured context object assembled at RDSG Stage 0. Contains the five relevancy dimensions (WHO/WHAT/WHEN/WHERE/WHY), activated domain holon references, and PEP scope definitions that govern the entire subsequent gateway traversal.
RDF	Resource Description Framework (W3C) — the universal holonic triple representation protocol (Subject–Predicate–Object). Knowledge graph storage, provenance, and lineage representation at L1–L4 of the KIH.
RDSG	Requirements-Driven Semantic Gateway — the Level 3 holarchic traversal and composition protocol. Six-stage flow (RCB Assembly → Enforcement Engine → Circuit Breaker → Compiled AI Directive → Simplex Gate → Service Facade) that assembles semantic authority from domain holons and delivers governed context to AI agents.

Acronym / Term	Definition & Architecture Role
R2RML	RDB to RDF Mapping Language — the standard used in the KIH STG environment to generate holonic representations of legacy relational data assets, enabling bottom-up holonic onboarding without data migration.
SHACL	Shapes Constraint Language (W3C) — the knowledge holon invariant enforcement language. SHACL constraints define what valid instances of a knowledge holon look like. Evaluated at RDSG Step 1 and in the inbound admission pipeline.
SLO	Service Level Objective — the SRE-style target values applied to AI knowledge services in the KIH. SLOs cover insight accuracy, latency, trust fidelity, data currency, and agent availability.
SPARQL	SPARQL Protocol and RDF Query Language (W3C) — the primary query language for holarchic traversal at L2/L3. Enables multi-hop relationship queries, impact analysis, and provenance chain traversal across the knowledge graph.
SRE	Site Reliability Engineering — the operational framework applied to AI knowledge service delivery in the KIH. Treats insight delivery as a critical operational service with formal SLOs, automated failover, and continuous degradation monitoring.
STAMP	Systems-Theoretic Accident Model and Processes (MIT Leveson) — the safety analysis framework whose UCA (Unsafe Control Action) enumeration is integrated into RDSG Step 4 for safety boundary enforcement at the gateway level.
STG	Staging Environment — the holonic composition lifecycle stage where top-down (core ontology + domain vocabularies) and bottom-up (legacy schema ontologies via R2RML) knowledge artifacts are reconciled. Cross-domain ontology alignment and mapping ontology creation occur in STG.
STPA	Systems-Theoretic Process Analysis — the MIT-derived hazard analysis methodology used to enumerate UCAs (Unsafe Control Actions) that the RDSG Simplex Action Gate enforces against.
SUDA	Sense → Understand → Decide → Act — the cognitive traversal protocol for all KIH interactions. Sense: 5D context tagging; Understand: L2 domain holon activation; Decide: L3 RDSG Gateway traversal; Act: governed domain agent execution.
SWRL	Semantic Web Rule Language (W3C) — the knowledge holon inference rule language. Applied at RDSG Step 2 for temporal reasoning and domain-specific inference logic. Extended with temporal rules for freshness enforcement.
UCA	Unsafe Control Action — enumerated at RDSG Step 4 via STPA/STAMP analysis. A control action that, in a given context, may lead to a hazardous state. The RDSG Simplex Action Gate prevents UCAs from reaching external actuators without human authorisation.
URI	Uniform Resource Identifier — the persistent holonic identity assigned to every knowledge holon. Together with IRI (internationalised form), URI is the mechanism by which identity is preserved across all contexts and lifecycle stages.
Value-Drift Coefficient	A numerical measure of the distance between an AI agent's current objective function and the original human intent encoded in its governing domain ontology. Holonic divergence metric monitored continuously; Domain Governance Team review triggered above 0.15.

The Value Enablement Group, LLC

Jeffrey Wallk · Senior Enterprise Architect & Knowledge Engineering Leader

KIH-ARCH-001-v1.0 · April 2026 · CONFIDENTIAL

Holonic Semantic Governance Architecture v2.0 · RDSG v2.0 · MTBH Trust Metrology · MCP Protocol